

STEPGEN · DESIGN DECISION RECORD

StepGen recommendations accepted

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DOCUMENT CONTROL

Document ID	SGN-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepgen		
Source	docs/decisions/0002-recommendations-accepted.md @ 8182ad3		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 acceptance of the TRL 3 review recommendations, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 5); items 6 and 7 remain proposed, awaiting Amish

■ Context

The TRL 3 review note (docs/REVIEW.md , session "2026-09-25: TRL 3") listed seven items as "Proposed, awaiting Amish", and SGN-DDR-001 carried them as its open item 15. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item that had a recommendation is therefore decided as recommended. Items with no recommendation stay open. TRL 3 remains the hard stop; TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 3 session) and SGN-CAL-001 v0.1. Where several options were offered, the recommended option is the decision.

■ Decision

Table 1. Items decided on 2026-09-25. Each is "Decided by Amish, 2026-09-25: go with recommendation".

#	Item	Decision	What changed in the repo
1	Cost overrun (R12)	Option (a): keep <code>budget_usd</code> at \$650 and make the salvage route (a used walking-pad treadmill for items 2 and 3, a donor 20 in bike for items 7 and 9) the reference build, with new parts as the fallback	<code>budget_usd</code> unchanged at \$650. R12 in SGN-REQ-001 restated to apply to the reference build. <code>docs/04-calcs/sizing.py</code> now checks R12 on the reference build: not met at \$713 becomes met on paper at \$546 (all-new fallback \$721 after item 3). <code>bom/bom-notes.md</code> shows both totals
2	Belt length (R1 against R10)	Option (a): keep the 1.05 m roller pitch and 2.35 m length; revisit after co-design shows the riders' heights and pace	No geometry change. R1 stays at risk, now marked as held until co-design in SGN-REQ-001, SGN-PRC-001 and SGN-CAL-001
3	Deck rails	Change from 50 x 25 x 2 mm to 60 x 30 x 2 mm RHS	<code>cad/src/model.py</code> (<code>rail_w</code> 25 to 30, <code>rail_h</code> 50 to 60), STEP and STL re-exported, SGN-DWG-001 to Rev P2, media regenerated. Weld stress range 37.6 to 25.5 MPa (limit 38.8 MPa); static factor 2.0 to 3.0. Frame 11.4 to 12.5 kg; vehicle 34.0 to 35.0 kg, 36.8 to 37.9 kg with the pack (R10 margin 3.2 to 2.1 kg, still at risk). Hill power 233 to 235 W (R6 margin 7 to 6 %). BOM item 1 \$75 to \$83
4	Key switch in the INTERLOCK loop	Option (a): a key switch in series with the 10 kΩ coding resistor; off opens the loop and sleeps the pack, on wakes it; also a simple immobiliser; used at a standstill only	Already modelled and priced (BOM items 10 and 13); SGN-PRC-001 and SGN-CAL-001 now mark it as decided instead of proposed
5	Flag to the SwapCell project	Raise with SwapCell: v0.3 does not say whether a pack in legacy discharge moves to mode 2 when a heartbeat arrives	Listed under "Cross-repo actions" in <code>docs/REVIEW.md</code> . The SwapCell repo is not edited from here

Items that remain open

These had no recommendation and stay **Proposed, awaiting Amish**:

- **6. First user group.** Non-cycling adults, older adults or commuters (SGN-DDR-001 item 13).
- **7. First target country** for the legal category (SGN-DDR-001 item 14).

Consequences

- SGN-PRB-001, SGN-PRC-001 and SGN-REQ-001 move from 0.4 to 0.5; SGN-CAL-001 moves from 0.1 to 0.2; SGN-DDR-001 moves from 0.1 to 0.2 to mark its item 15 as decided.
- Requirement status: R12 moves from not met to met on paper. No requirement is now not met. R1, R6 and R10 remain at risk; R9 and R13 remain not verifiable at TRL 3.
- The deck-rail weld fatigue concern in the TRL 3 review is closed on paper; it still needs a test at TRL 4.
- The salvage prices (\$40 for a walking-pad treadmill, \$50 for a donor bike) are estimates; R12 now depends on them.
- `trl` and `trl_target` stay at 3. No TRL 4 work (test article, test plan, build log, firmware) is started; it is on hold by Amish's instruction.